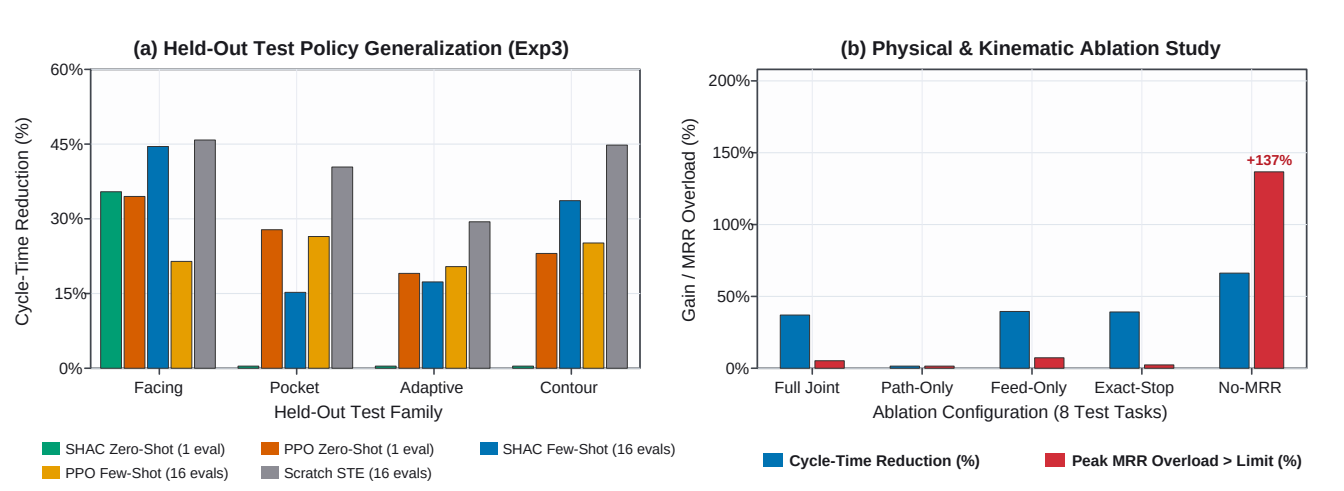
(a) Held-Out Test Policy Generalization (Exp3)
Cycle-Time Reduction (%)
60%
45%
30%
15%
0%
Facing
Pocket
Adaptive
Contour
Held-Out Test Family
SHAC Zero-Shot (1 eval)
PPO Zero-Shot (1 eval)
SHAC Few-Shot (16 evals)
PPO Few-Shot (16 evals)
Scratch STE (16 evals)
(b) Physical & Kinematic Ablation Study
Gain / MRR Overload (%)
200%
150%
100%
50%
0%
+137%
Full Joint
Path-Only
Feed-Only
Exact-Stop
No-MRR
Ablation Configuration (8 Test Tasks)
Cycle-Time Reduction (%)
Peak MRR Overload > Limit (%)